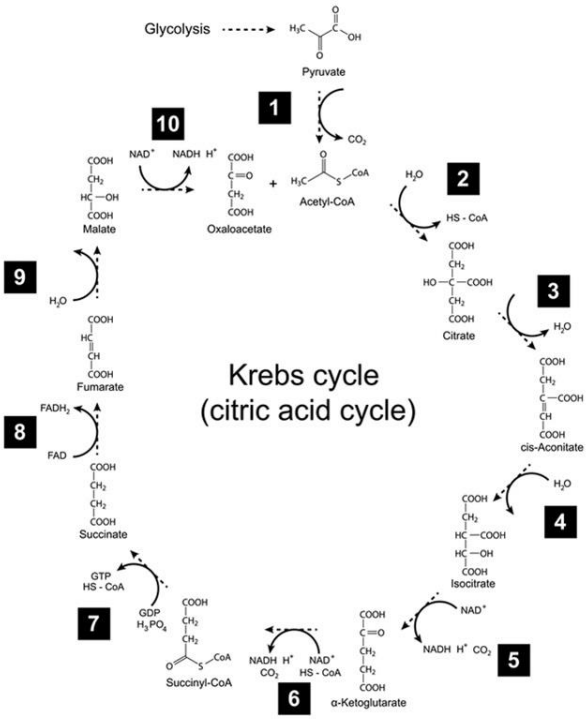
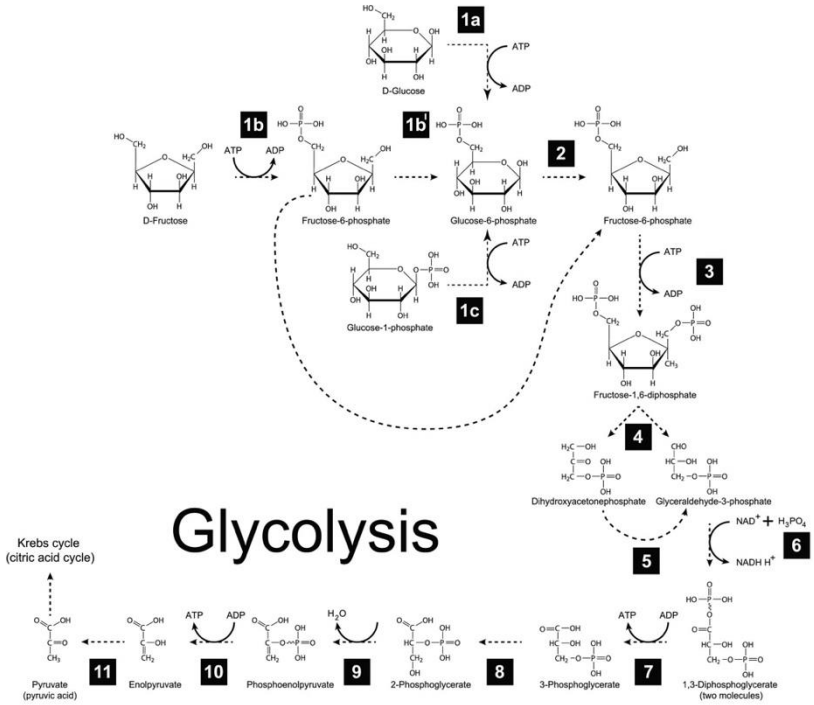
Biochemistry Cume Question

Understanding metabolic pathways is essential in medicine and biotechnology, as it provides insight into diseases caused by metabolic defects and informs the development of therapeutic interventions.

Question: (100 pts)
Trace the fate of carbon atom number 1 (C2, labeled with ^{14}C) in a glucose molecule as it proceeds through the citric acid cycle (TCA cycle, Krebs cycle) and is ultimately converted to aspartate (Asp) via transamination. Provide a step-by-step account of the relevant reactions and intermediates in both the TCA cycle and the transamination process. Clearly indicate the position of the ^{14}C label in each intermediate.

Instructions:

- Show how the ^{14}C-labeled carbon atom moves through each step of the citric acid cycle.
- Identify the step at which the labeled carbon atom is incorporated into aspartate during the transamination reaction.